



Express Middlewares

1. What is middleware
2. Custom Middlewares
3. Built-in Middleware

- Functions that get executed in the middle of the sequence of request processing.
- Used to modify the incoming request, read the request cookies or data, modify the response etc.
- Arguments for the middleware functions are request(object), response(object) and next function
- Scenarios where middleware is required
 - Generating log of the requests
 - Reading request body or cookies
 - Compressing responses etc
- Functions of middleware
 - Change request or response objects
 - End the request processing if needed
 - Call next middleware in sequence

- Syntax of Middleware

```
app.use('route path',function(request, response, next){  
    //middleware logic  
})
```
- Types of Middleware functions
 - i. Built-in
 - ii. Custom
 - iii. Third-party
- Order of execution
 - get executed exactly in the same sequence the way they are written
 - Call to next() is very crucial

- Created by the developer to add custom functionality to the Express.js application
- Middleware can be bound to app object by using the `app.use()` and `app.METHOD()` (any http method) functions
- First parameter in `use()` method is optional if not specified, it creates general route.
- Middlewares can be associated with specific request URL.

- Few functions are bundled with Express
- Main example of built-in middleware functions
 - `express.static`
 - `express.json`.
- Using `express.static`
 - `express.static` built-in middleware function to serve static files such as images, CSS files, and JavaScript files
 - We can define multiple static assets directories
 - Express will look for the static files in the directories mentioned in the static function call
 - No need to mention the file of the files in the response

server.js

- Using `express.json`
 - This method is used to parse the incoming requests with JSON data
 - It automatically converts the incoming JSON data into a JavaScript object, making it accessible within your Express routes.
 - The parsed JavaScript object is then made available as the `req.body` object